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Research interests

LLM post-training and evaluation, retrieval-augmented generation, agentic systems, and integration with graph-based methods.

Research experience

Institute for Logic, Language and Computation, UvA

Promoter: Khalil Sima'an, Supervisor: Jelke Bloem October 2022 - Present
LLM post-training and evaluation, and the coordination and alignment between language models and graph models.

University of Chinese Academy of Sciences (UCAS)

Supervisor: Junyong Fang October 2014 - July 2017
Analyzing multi-band remote sensing images using deep learning and computer vision techniques.

Medical College, Peking University

Part-time Researcher 2018 - 2019
Information retrieval and question answering for Chinese medicine literature

Industry experience

Kunlun Digital Technology Co.,Ltd. (or CNPC Richfit)

Senior Machine Learning Researcher January 2021 – October 2022
Team Leader in Conversation Service Group. Responsible for building an automated customer question-answering system, leading the team, and managing projects. Service robots equipped with the QA system [\[Video\]](#) [\[Description\]](#).

Tsinghua Tongfang Co., Ltd.

Machine Learning Researcher July 2017 – January 2021
Team Leader in Machine Learning Group. Responsible for research on natural language processing, knowledge graphs, and text mining algorithms; applying these algorithms in projects; and leading the team while managing projects.

Selected Publications

[Fusion Training for Mathematical Generalization in Large Language Models \(ACL 2026 SRW\)](#)

Congfeng Cao, Pengyu Zhang, Jelke Bloem.

Contribution: Reveals thinking and non-thinking training modes interfere, and mixed training gives the best balance.

Technical: Qwen3-4B base model, Supervised Fine-Tuning (SFT), GSM8K, Open-MathReasoning, thinking/non-thinking data ratios, T-NT/NT-T/Mix schedules.

[Graph-TempCZ: A Graph Representation of Software Mentions for Predicting Software Usage in Scientific Publications. \(LREC 2026\)](#)

Congfeng Cao, Pengyu Zhang, Jelke Bloem.

Contribution: Introduces Graph-TempCZ, a temporal graph representation of software mentions, and formulates software usage prediction in scientific publications as a link prediction task.

Technical: Sentence-BERT, XGBoost, GraphSAGE, temporal graph evaluation, CZ Software Mentions, and Europe PMC metadata.

How Aligned Are Unimodal Language and Graph Encodings of Chemical Molecules? (IJCNLP-AACL Main 2025)

Congfeng Cao, Zhi Zhang, Jelke Bloem, Khalil Sima'an.

Contribution: Studies whether molecular graph and language encoders are inherently aligned before force-alignment training, and introduces the U2A benchmark to evaluate this alignment.

Technical: MolCLR, MolFormer, SciBERT, SentenceBERT, GPT, Debaised CKA, U2A benchmark, ChEBI-20, RDKit, and MAYGEN.

Under Review

SoftwareCiter: Agentic Generation of Verifiable Software Citations with a Metadata-Aware Benchmark

Congfeng Cao, Yixian Shen, Kayle Verhiel, Jelke Bloem.

Contribution: Proposes a multi-agent framework for generating complete and verifiable software citations from incomplete software mentions in publications

Technical: LLM agents, publication parsing, metadata extraction, web searching, FORCE11 citation principle, and citation metadata verification.

Molecule Generation through Reasoning with Large Language Models

Congfeng Cao, Di Wu, Jiahong Huang, Jelke Bloem, Khalil Sima'an.

Contribution: Frames molecule generation as reasoning-based SMILES generation and introduces MolCoT-4K and MolGeneration.

Technical: Qwen2.5-7B base model, curated CoT data, SFT, GRPO RL, RDKit validity checking, exact-match rewards, and Pass@N evaluation.

Rethinking Feature Augmentation in Graph Contrastive Learning

Congfeng Cao, Pengyu Zhang, Zeyu Zhang, Jelke Bloem.

Contribution: Introduces Random Feature Masking (RFM), a simple graph contrastive learning augmentation that improves node classification performance and robustness, especially at high masking rates.

Technical: Graph contrastive learning, feature augmentation, GRACE, GraphCL, CCA-SSG, node classification, and uniformity analysis.

A Graph-Based Analysis of Software Usage in Scientific Publications

Congfeng Cao, Na Li, Kristy James, Jelke Bloem.

Contribution: Shows software forms structured, domain-specific ecosystems rather than isolated frequency patterns.

Technical: Graph-CZ, Software-CZ, Publication-CZ, temporal analysis, community detection, and cross-community interaction analysis.

Grants

NWO Compute Projects

2024, 2025, 2026

UCAS Academic Scholarship

2015, 2016, 2017

Education

University of Amsterdam

Amsterdam, Netherlands

PhD in Computer Science

October 2022 – Present

University of Chinese Academy of Sciences

Beijing, China

MEng in Signal and Information Processing

October 2014 – July 2017

Shandong Agricultural University

Taian, China

BEng in Computer Science, BEng in Remote Sensing October 2010 – July 2014

Presentations

ACL 2026, LREC 2026, IJCNLP-AAACL 2025, EMNLP 2025, CLIN 2025, CLIN 2024, ELLIS 2024.

Course & Training

[Knowledge Graphs](#) (2026), [IBM RAG and Agentic AI Professional Certificate](#) (2025), [Agentic AI](#) (2025), [Fine-tuning & RL for LLMs](#) (2025), [Generative AI with Large Language Models](#) (2025), [Machine Learning and Optimisation](#) (2024), [LxMLS](#) (2024), [LOGML](#) (2025), [Human-Centred AI](#) (2023)

Skills

Programming

Proficient in: Python, PyTorch, TensorFlow. Familiar with: C/C++, Java.

Database: SQL (MySQL, Oracle, PostgreSQL), NoSQL (Neo4J, MongoDB)

Languages

Chinese (native), English (work language), Dutch (Beginner)